Gravity above the surface

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Use distance from a planet's center to compare fields at altitude.

SUCCESS CRITERIA - I can...

1. Use r=R+h rather than altitude alone.
2. Compare fields with the inverse-square relation.
3. Explain why orbit is not a zero-gravity region.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use "altitude" and "radius" to compare your model or result. Sentence starters are optional.

altitude: height above a reference surface

radius: distance from the center to the edge of a circle or sphere

field: a description of an influence at each location

01 Bring back an earlier idea

For fixed masses, what separation multiplier makes gravity 1/9 of its original value? What multiplier is needed if both masses also become 5 times as large?

Word help: separation = distance between the specified objects or centers

Sentence starter: I notice … . This supports … because … .

Marzano check - circle: ① ② ③

02 Explain a worked example

Reference: $r = R + h$; $g_h/g_{surface} = [R/(R + h)]^2$.
R is planet radius; h is altitude. g_h is field strength at altitude h.

Worked reasoning: At altitude h=2 R above a planet's surface, center distance is r=R+h=3 R. A surface field of 18 N/kg becomes 18/9=2 N/kg.

Your turn: Draw the radius and altitude as different line segments.
Mark the full center-to-object distance.

Sentence starter: Both … . They differ in … because … .

Marzano check - circle: ① ② ③

03 Represent and interpret

A satellite is one planet-radius above the surface.
Label R, h and r on a sketch. What is r in units of R?

Sentence starter: My design shows … . It meets the requirement by … .

Marzano check - circle: ① ② ③

04 Solve and justify

For Earth use$R=6.4\times10^6$ m, $g_{surface}=9.8$ N/kg.
Find g at $h=4.0\times10^5$ m and at h=R.

Sentence starter: I found … [unit]. This value means … in this situation.

Marzano check - circle: ① ② ③

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Continue down each column.

Sentence starters are optional. You may plan in any language.

05 Analyze the reasoning

Explain why an astronaut at modest orbital altitude can float even though gravity is only slightly weaker there.

Sentence starter: Under the assumption ..., the evidence ... supports ... because ... .

Marzano check - circle: 1 2 3

06 Choose a method

A 4 kg box has μ_s=0.50 and μ_k=0.40 on a level floor; g=9.8.
Compare pushes of 8 N and 24 N, starting from rest.
Find actual friction and acceleration.

Word help: friction = a contact force opposing relative sliding or its tendency

Sentence starter: Both ... . They differ in ... because ... .

Marzano check - circle: 1 2 3

07 Exit check

At what altitude in units ofR is field one quarter of surface value? At what altitude is it one ninth?

Sentence starter: I notice ... . This supports ... because ... .

Marzano check - circle: 1 2 3

08 STRETCH: Apply it another way

Find the altitude h/R where a spherical planet's field is 10% of the surface value.
Explain why h/R is not √10.

Sentence starter: My design shows ... . It meets the requirement by ... .

Marzano check - circle: 1 2 3